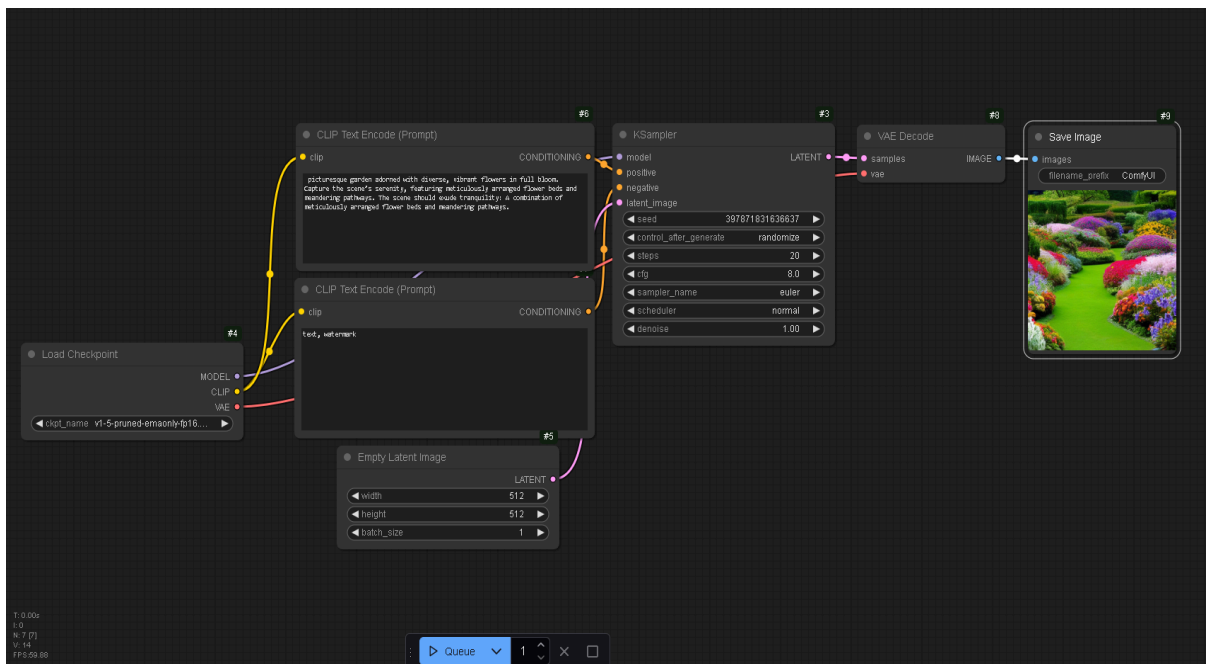


CHAPTER 3

Proposed Methodology

3.1 SystemDesign



The provided workflow showcases an image generation pipeline using **Stable Diffusion** within **ComfyUI**, a node-based interface for AI-based image synthesis. Below is a breakdown of each node in the diagram:

1 Load Checkpoint (a.k.a. "The AI Brain")

Purpose: Loads the **pre-trained Stable Diffusion model**—the "brain" that knows how to generate images.

Parameters Used:

- **Model Checkpoint (v1-5-pruned-emaonly-fp16...)** → Determines which version of Stable Diffusion we're using.
- **CLIP** → Handles text-to-image understanding.
- **VAE** → Converts raw AI data into human-readable images.

What We're Training Here:

This is a **pre-trained** model, meaning it already knows how to generate images. We're just **using** it.

2 CLIP Text Encode (Prompt)

Purpose: Converts our text description into something the AI can understand.

Parameters Used:

- **Text Prompt:** "Picturesque garden adorned with vibrant flowers..."
 - This is the **main instruction** that tells AI what to create.

What We're Training Here:

- We're **guiding** the AI using text, but **not training a new model**.
- The AI follows the prompt, but results **can vary based on model knowledge**.

3 CLIP Text Encode (Negative Prompt)

Purpose: Stops the AI from generating unwanted elements.

Parameters Used:

- **Negative Prompt:** "text, watermark"
 - This prevents unwanted text, watermarks, or extra junk in our image.

What We're Training Here:

- We're **filtering out** things we don't want, ensuring cleaner results.

4 Empty Latent Image (a.k.a. "The Blank Canvas")

Purpose: Creates an **empty latent space** where our AI-generated image will be drawn.

Parameters Used:

- **Width:** 512px
- **Height:** 512px
- **Batch Size:** 1 (How many images we generate at once)

What We're Training Here:

- Nothing, just defining **image dimensions** before generation starts.

5 KSampler (a.k.a. "The AI Artist at Work")

Purpose: This is the **core** of the Stable Diffusion pipeline. It takes all the previous inputs and **actually generates** the image.

Parameters Used:

- **Seed:** 39781781363637
 - Determines **randomness**. Change it, and you get a different image.
- **Control After Generate:** **randomize**
 - Makes results less predictable (good for creative freedom).
- **Steps:** 20
 - More steps = **better detail**, but slower generation.
- **CFG Scale:** 8.0
 - Controls how strictly the AI follows the prompt.
 - **Lower (~4-6)** → AI gets creative.
 - **Higher (~10-12)** → AI sticks very close to the prompt.
- **Sampler Name:** **Euler**
 - Different samplers affect how the image is **refined step-by-step**.

- **Scheduler:** Normal
 - Affects how diffusion noise is processed.
- **Denoise Strength:** 1.00
 - Affects how much noise is removed from the image. Higher = sharper.

What We're Training Here:

- We're **not training a new model**, but we're tweaking settings to control how the AI refines the image.

6 VAE Decode (a.k.a. "Bringing the Image to Life")

Purpose: Converts the **latent (fuzzy AI representation)** into a clear, **final image**.

Parameters Used:

- **Samples** (Input from KSampler) → The AI-generated raw image data.
- **VAE Model** → Decodes and refines the image.

What We're Training Here:

- Again, no **training**, just converting **AI magic into human-readable visuals**.

7 Save Image (a.k.a. "The Final Touch")

Purpose: Saves our stunning AI-generated image so we can use it anywhere.

Parameters Used:

- **Filename Prefix:** "ComfyUI" (You can change this).

What We're Training Here:

- Just **saving** the finished image.

3.2 Requirement Specification

3.2.1 Hardware Requirements:

Processor: Minimum Intel i5 / Ryzen 5, Recommended Intel i7+ / Ryzen 7+

GPU: Minimum NVIDIA GTX 1660 (6GB VRAM), Recommended RTX 3060 (12GB) or higher

RAM: Minimum 8GB, Recommended 16GB or more

Storage: At least 20GB free space (for models & image storage)

OS: Windows 10/11, Linux (Ubuntu), macOS (M1/M2)

3.2.2 Software Requirements:

Operating System: Windows, Linux, or macOS

Stable Diffusion Model: v1.5 / v2.1 / SDXL

ComfyUI: Node-based interface for Stable Diffusion

Python: Version 3.8+ required for running scripts

CUDA: Required for NVIDIA GPU acceleration

GitHub: Version control and collaboration

Torch & torchvision: Essential libraries for AI model execution

WebUI (Optional): Alternative UI for easier access

Image Editing Tools: Photoshop, GIMP for post-processing

CHAPTER 4

Implementation and Result

4.1 Snap Shots of Result:

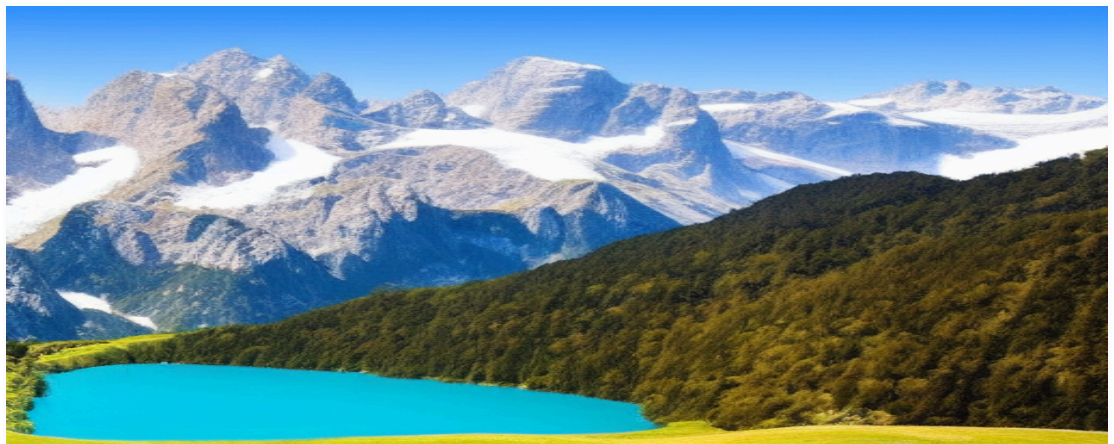
- a. picturesque garden adorned with diverse, vibrant flowers in full bloom. Capture the scene's serenity, featuring meticulously arranged flower beds and meandering pathways. The scene should exude tranquility: A combination of meticulously arranged flower beds and meandering pathways.



- b. Generate a surreal image of an abstract and mechanical garden, where metallic flowers adorned with gears serve as petals and robotic vines climb imaginary structures.



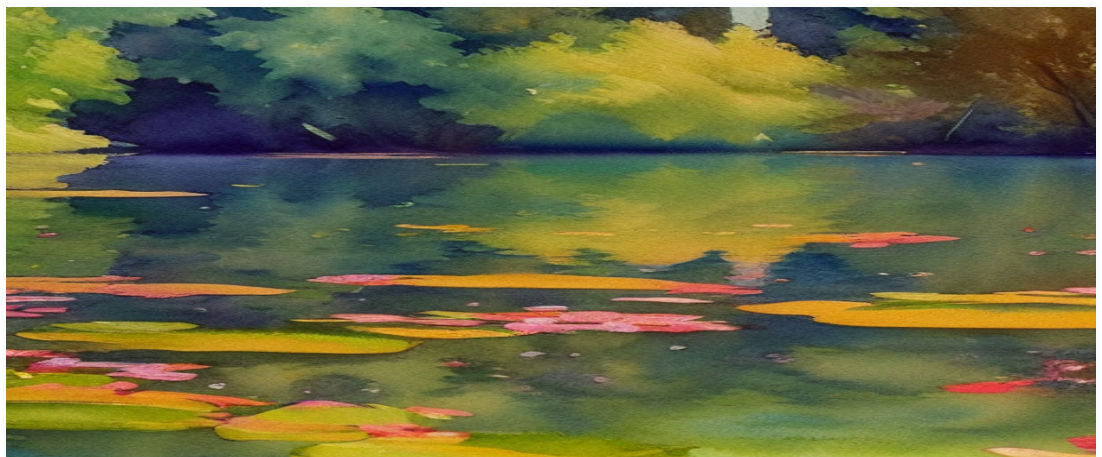
- c. Generate an image capturing the stunning landscapes of the Alps. Showcase towering, snow-capped peaks, lush green meadows, and crystal-clear alpine lakes. Include quaint villages nestled in the valleys



- d. Generate a series of realistic photographs that capture everyday moments in a genuine and relatable manner. Picture scenes of people going about their daily lives, candid expressions, and the beauty found in ordinary details



- f. Generate an impressionist-style watercolor oil painting of a tranquil pond. Use brushstrokes to blend colors, capture the play of light and color, and put in a warm glow of dappled sunlight



- g. Generate a presentation slide on sales opportunities in a particular geography. Include a map highlighting key regions and concise bullet points outlining market potential, target demographics, and potential growth areas.



- h. Generate options of emblems for a prestigious college, with symbols such as a shield, an open book, and a pen. Arrange these elements in a good color scheme, creating a sense of knowledge, tradition, and enlightenment.



i.

- j. A detailed portrait of a mythical creature, like a dragon or a mermaid



k. An astronaut exploring a vibrant alien planet.



l.

4.2 GitHub Link : <https://github.com/Kdayma00/stable-diffusion>